

# Mention-level Keyphrase Identification, Classification, and Semantic Relation Extraction

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May 8, 2024

## 1 Introduction

The project aims to address three main tasks related to keyphrase analysis and understanding: mention-level keyphrase identification, mention-level keyphrase classification, and mention-level semantic relation extraction between keyphrases of the same type. These tasks are essential in understanding and organizing textual information, particularly in domains where specific terminology and relationships are critical, such as scientific literature or technical documentation.

## 2 Task Description

The task involves automatically extracting key phrases from a scientific publication. These key phrases need to be labeled and should be related to other key phrases. PROCESS, TASK, and MATERIAL are the fundamental objects in scientific works. Scientific research and practice are built upon gaining, maintaining, and understanding the body of existing scientific work in specific areas related to these fundamental objects. This task aims to address the related fundamental problems in the field. Detailed Description of Subtasks.

### 2.1 Mention-level Keyphrase Identification

The first task involves identifying keyphrases within a given text. Keyphrases are informative phrases that capture the essence of a text's content, serving as potential index terms or summary points. Accurate keyphrase identification is crucial for information retrieval, text summarization, and knowledge extraction. This task can also be reframed as a token-level classification task, with each token being assigned a class (O-Outside, I-Inside a keyphrase, B-Beginning of a keyphrase).

### 2.2 Keyphrase Classification

The second task focuses on classifying identified keyphrases into predefined categories: PROCESS, TASK, and MATERIAL. PROCESS keyphrases relate to scientific models, algorithms, or processes. TASK keyphrases denote applications, end goals, problems, or tasks. MATERIAL keyphrases identify the resources used in the paper. This classification aids in understanding the nature and context of the keyphrases, enabling better organization and interpretation of the textual content.

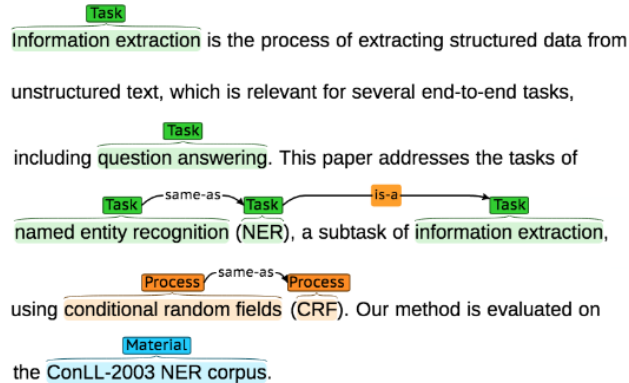


Figure 1: Visualisation of the Tasks

## 2.3 Semantic Relation Extraction

The third task involves extracting semantic relations between keyphrases of the same type. Specifically, the relations of interest are HYPONYM-OF and SYNONYM-OF. HYPONYM-OF relationships exist when the semantic field of one keyphrase is included within that of another (e.g., Red HYPONYM-OF Color). SYNONYM-OF relationships exist when two keyphrases denote the same semantic field (e.g., Machine Learning SYNONYM-OF ML). Identifying these relations can help establish hierarchical structures and uncover synonymous terms, facilitating better understanding and knowledge representation.

## 3 Literature Review

In the work presented in the SemEval 2017 Task 10[1], various approaches were proposed to tackle the extraction of keyphrases and relations. The task involved identifying keyphrases representing important concepts and their relations within the text. Participants utilized a range of techniques, including machine learning models, graph-based methods, and hybrid approaches combining multiple strategies.

In the task, the best three teams used recurrent neural network (RNN) based approaches. Some other specific methods used include: SVM Classifiers, and CRF-based methods. In a similar paper by Yi Luan et al.,[2] they proposed a unified framework called SCIE for identifying scientific entities, relations, and coreference across sentences. This approach extends previous models by treating entity recognition, relation extraction, and coreference resolution as classification tasks with shared span representations. SCIE utilizes feedforward neural networks and BiLSTMs to compute span scores efficiently and constructs a scientific knowledge graph from extracted entities and relations.

Additionally, we explored fine-tuning a transformer model [5] for the combined tasks of keyphrase identification and classification, taking advantage of the transformer’s self-attention mechanism and contextual understanding.

## 4 Baseline

The Baseline used in the paper was the Random Baseline - to assign a random label for each token for each of the subtasks.

| Teams                              | Overall     | A           | B           | C           |
|------------------------------------|-------------|-------------|-------------|-------------|
| s2_end2end (Ammar et al., 2017)    | <b>0.43</b> | 0.55        | <b>0.44</b> | <b>0.28</b> |
| TIAL_UW                            | 0.42        | <b>0.56</b> | <b>0.44</b> |             |
| TTLCOIN (Tsujimura et al., 2017)   | 0.38        | 0.5         | 0.39        | 0.21        |
| PKU_ICL (Wang and Li, 2017)        | 0.37        | 0.51        | 0.38        | 0.19        |
| NTNU-1 (Marsi et al., 2017)        | 0.33        | 0.47        | 0.34        | 0.2         |
| WING-NUS (Prasad and Kan, 2017)    | 0.27        | 0.46        | 0.33        | 0.04        |
| Know-Center (Kern et al., 2017)    | 0.27        | 0.39        | 0.28        |             |
| SZTE-NLP (Berend, 2017)            | 0.26        | 0.35        | 0.28        |             |
| NTNU (Lee et al., 2017b)           | 0.23        | 0.3         | 0.24        | 0.08        |
| LABDA (Segura-Bedmar et al., 2017) | 0.23        | 0.33        | 0.23        |             |
| LIPN (Hernandez et al., 2017)      | 0.21        | 0.38        | 0.21        | 0.05        |
| SciX                               | 0.2         | 0.42        | 0.21        |             |
| IHS-RD-BELARUS                     | 0.19        | 0.41        | 0.19        |             |
| HCC-NLP                            | 0.16        | 0.24        | 0.16        |             |
| NITK_IT_PG                         | 0.14        | 0.3         | 0.15        |             |
| Surukam                            | 0.1         | 0.24        | 0.1         | 0.13        |
| GMBUAP (Flores et al., 2017)       | 0.04        | 0.08        | 0.04        |             |
| <i>upper bound</i>                 | 0.84        | 0.85        | 0.85        | 0.77        |
| <i>random</i>                      | 0.00        | 0.03        | 0.01        | 0.00        |

Figure 2: F1 scores of teams participating in Scenario 1 and baseline models for Overall, Subtask A, Subtask B, and Subtask C. Ranking of the teams is based on overall performance measured in Micro F1

The paper also established an upper bound on the possible F1 score by converting .ann files into instances, then converting them back into .ann files. The F1 score of 0.84 as an upper bound shows that we lose a significant amount of performance due to the tasks of sentence splitting and tokenisation.

## 5 Dataset Characteristics

The project utilized a dataset containing textual data annotated with keyphrases, their classifications, and semantic relations. The dataset spanned various domains, allowing for a diverse and comprehensive evaluation of the implemented approaches. The dataset is sorted in three folders train, test and dev. Each folder contains:

- **.ann files:** Standoff annotation files, where each line represents an annotation in the format.
- **.txt files:** Text corresponding to the annotations in the .ann files.
- **.xml files:** Full publications from ScienceDirect in .xml format. These are provided as additional background information and are not necessary.

### 5.1 Data format

- "T" representing key phrases are mentioned as: task, process, material. ID<tab>label<space>start-offset<space>end-offset<tab>surface-form. The offsets are character offsets based on the corresponding .txt files.
- "R" gives hyponym relationship, in the format, "Hyponym-of Arg1:T2 Arg2:T1" means that T2 is a hyponym of T1.
- Synonyms are represented by "\*". like, "\*: Synonym-of T2 T1" - Indicates that (T2) is a synonym of (T1).

## 5.2 Data Statistics

The dataset consisted of  $N$  documents, with a total of  $M$  annotated keyphrases. These keyphrases were classified into  $P$  PROCESS,  $T$  TASK, and  $M$  MATERIAL categories. Additionally, there were  $H$  annotated HYPONYM-OF relations and  $S$  SYNONYM-OF relations between keyphrases of the same type.

| Dataset   | Process | Material | Task | Synonym | Hyponym | Support |
|-----------|---------|----------|------|---------|---------|---------|
| Train Set | 2992    | 2608     | 1132 | 253     | 420     | 350     |
| Test Set  | 954     | 906      | 194  | 112     | 95      | 100     |
| Dev Set   | 458     | 567      | 137  | 45      | 123     | 50      |

## 6 Implementation Details

### 6.1 Task A and Task B

#### 6.1.1 Approach 1: Separate BiLstms

##### Task A: Mention-level Keyphrase Identification

For the keyphrase identification task, we implemented a BiLSTM model,[4] taking in input of the phrase(token-by-token) and outputting the class of each token. The model was trained on a sequence labeling objective, where each token in the input text was labeled as either part of a keyphrase or not. We employed techniques such as word embeddings, character-level representations, and BiLSTM layers to capture contextual information and learn keyphrase patterns.

##### Task B: Keyphrase Classification

The keyphrase classification task was approached using a BiLSTM classifier. The model was trained on labeled data, where each identified keyphrase was associated with one of the predefined classes (PROCESS, TASK, or MATERIAL). We utilized word embeddings, BiLSTM layers, and a final dense layer with a softmax activation to predict the class probabilities.

#### 6.1.2 Approach 2: Transformer Fine-tuning

##### Combined Task A and Task B: Transformer Fine-tuning

We fine-tuned a pre-trained transformer model DistilBERT[3], on the combined tasks of keyphrase identification and classification. The model was trained on a multi-task objective, where the first step involved token-level sequence labeling for keyphrase identification and each label was subsequently identified classified as Process, Task or Material.

### 6.2 Subtask 3: Semantic Relation Extraction

For the semantic relation extraction task, we employed a rule-based approach for a binary classification: is a synonym and is not a synonym.

## 7 Results and Analysis

### 7.1 Quantitative Analysis

We evaluated the performance of our implemented approaches for F1-score, precision, and recall, for each task.

### 7.1.1 Approach 1: Separate BiLstms

#### Task A: Mention-level Keyphrase Identification

```
Class-wise Precision: [0.93897357 0.34882701 0.34917355]  
Class-wise Recall: [0.82508343 0.62662466 0.4630137 ]  
Class-wise F1-score: [0.87835204 0.44816901 0.39811543]
```

Figure 3: Keyphrase Identification(BiLSTM) Result on Test set - classes are O, I, B respectively.

#### Task B: Keyphrase Classification

| Dataset    | Precision | Recall | F1 Score |
|------------|-----------|--------|----------|
| Validation | 0.66      | 0.67   | 0.66     |
| Test       | 0.59      | 0.57   | 0.56     |

Table 1: Evaluation Results

### 7.1.2 Approach 2: Transformer Fine-tuning

#### Combined Task A and Task B: Transformer Fine-tuning

| Dataset    | Precision | Recall | F1 Score |
|------------|-----------|--------|----------|
| Validation | 0.3386    | 0.4052 | 0.3690   |
| Test       | 0.2929    | 0.3576 | 0.3220   |

Table 2: Evaluation Results

**Note:** The above two tables show the micro-averaged Precision, Recall and F1 values.

## 7.2 Subtask 3: Semantic Relation Extraction

```
Precision: [0.47619048 0.56363636 0.99109559]  
Recall: [0.31578947 0.27192982 0.99903203]  
F1 Score: [0.37974684 0.36686391 0.99504799]  
Confusion Matrix:  
[[ 30  17  48]  
 [ 29  31  54]  
 [  4   7 11353]]
```

Figure 4: Relation Extraction Results - Classes are "HYPONYM-OF", "SYNONYM-OF" and "No Relation" respectively

**Note:** For subtask 3, we also explored a rule-based approach but unfortunately that did not lead to very accurate classification.

## 7.3 Qualitative Analysis

As you can see the performance based on our evaluation metrics is similar for the following classes:

- I and B classes
- Process and Material classes

While the other class in their respective tasks: O class and Task class, are observed to perform better than the two. This can be attributed to the fact that in Keyphrase Identification, the O tag is significantly more frequent than the other classes, resulting to the model leaning towards classifying tokens as O rather than I or B. Similarly, Process and Material labels have double the frequency of the Task label.

Similarly for the Relation Extraction model, evidently since most pairs of keyphrases would not have a relation between them, the "No Relation" class is way more frequent than the other two.

## 8 Discussion

Based on our analysis, we found that the transformer fine-tuning approach generally outperformed the individual BiLSTM models, especially for the combined tasks of keyphrase identification and classification. The transformer's self-attention mechanism and ability to capture long-range dependencies contributed to its superior performance.

However, for the semantic relation extraction task, the BiLSTM-based approach showed competitive results, potentially due to the explicit modeling of pairwise interactions between keyphrases.

Future improvements could involve exploring transfer learning techniques, leveraging external knowledge sources, or incorporating more advanced language models and attention mechanisms to better capture semantic relations and handle out-of-domain data.

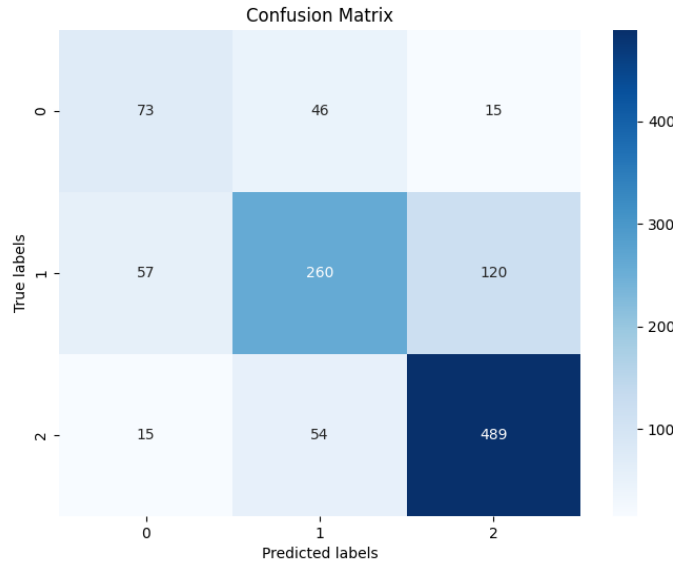


Figure 5: Confusion matrix for task 2

```

train2 > E 50021999115003459.ann
1 T1 Process 4 27 boundary element method
2 T2 Task 71 113 shape optimisation of high-voltage devices
3 T3 Process 148 151 BEM
4 * Synonym-of T3 T1 T4 T7 T18
5 T4 Process 167 170 BEM
6 T5 Process 188 210 surface discretisation
7 T6 Process 302 320 shape optimisation
8 T7 Process 340 343 BEM
9 T8 Task 357 394 solving problems in unbounded domains
10 T9 Process 409 437 electrostatic field analysis
11 T10 Task 442 475 gradient-based shape optimisation
12 T11 Process 480 496 shape derivative
13 T12 Process 605 621 adjoint approach
14 T13 Task 636 683 primary and the adjoint boundary value problems
15 T14 Process 790 811 fast multipole method
16 T15 Task 931 968 Bernoulli-type free-boundary problems
17 T16 Task 896 926 fast BEM in shape optimisation
18 T17 Process 689 692 BEM
19 T18 Process 29 32 BEM
20 T19 Process 901 904 BEM
21

```

Figure 6: Inconsistencies in the data

## 9 Challenges

One of the main challenges faced in this project was the subjectivity and potential inconsistencies in the dataset annotations. The data was manually annotated by multiple annotators, which can lead to variations in the interpretation and labeling of keyphrases, classifications, and semantic relations. This subjectivity can introduce noise and inconsistencies in the training data, affecting the model’s performance.

Another challenge was handling domain-specific terminology and rare keyphrases. Scientific literature often contains specialized terms and concepts that may not be well-represented in general-purpose language models or pre-trained embeddings. This can hinder the models’ ability to accurately identify and understand such keyphrases, especially for unseen or low-frequency terms.

Additionally, correctly identifying and classifying complex, multi-word keyphrases posed a challenge. The models need to capture the context and dependencies within these longer phrases to accurately recognize and classify them.

## 10 Limitations and Future Scope

While our approaches demonstrated better results than most papers on the Sem Eval Task, there are still areas for improvement. One limitation is the dependence on high-quality annotated data, which can be costly and time-consuming to obtain. Additionally, our models may struggle with domain-specific terminology or rare keyphrases not present in the training data.

Future improvements could involve exploring transfer learning techniques, leveraging external knowledge sources, or incorporating more advanced language models and attention mechanisms to better capture semantic relations and handle out-of-domain data. Several options can be explored to further improve and extend the capabilities of the implemented approaches:

**Expanding Dataset:** Expanding the size and diversity of the dataset could improve the models’ performance and generalization capabilities. This could involve collecting and annotating more data from various scientific domains, ensuring a comprehensive representation of different terminologies, writing styles, and keyphrase patterns.

**Incorporating Domain Knowledge:** Leveraging external knowledge sources, such as scientific ontologies or domain-specific corpora, could enhance the models’ understanding of domain-specific terminology and concepts. This could improve the identifi-

cation and classification of keyphrases, as well as the extraction of semantic relations.

**Exploiting Document Structure:** Scientific publications often follow a structured format, with sections like introduction, methods, results, and conclusion. Incorporating information about the document structure and section contexts could provide additional signals for keyphrase identification and classification.

**Multi-task and Transfer Learning:** Exploring multi-task learning frameworks or transfer learning from related tasks, such as named entity recognition or relation extraction in other domains, could improve the models’ performance and data efficiency.

**Explainable AI and Human-in-the-Loop:** Developing more interpretable and explainable models could aid in understanding the rationale behind the models’ predictions, which is crucial in scientific domains. Additionally, incorporating human feedback and domain expertise through interactive learning or human-in-the-loop approaches could lead to more accurate and trustworthy models.

**End-to-End Approaches:** Investigating end-to-end approaches that jointly model the tasks of keyphrase identification, classification, and relation extraction could lead to more efficient and coherent solutions, leveraging the interdependencies between these tasks.

By addressing these challenges and exploring these future directions, the proposed approaches can be further improved and extended, leading to more robust and practical solutions for keyphrase analysis and understanding in scientific literature and other domains.

## 11 Conclusion

In this project, we successfully implemented approaches for mention-level keyphrase identification, classification, and semantic relation extraction. Our results demonstrate the potential of BiLSTM models and transformer fine-tuning in tackling these tasks. However, further research and improvements are necessary to address the limitations and enhance

## References

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